

Hash table: effective algorithm

Advanced: Practice 2  

Which **three** of these criteria are the most important for a hashing algorithm to be considered effective and useful?

- ☐ Must be designed to minimise collisions
- ☐ Must ensure the resulting hash table is completely full, leaving no empty spots.
- ☐ Must generate the same hash value every time the same input key is given.
- ☐ Must aim to spread the generated hash values evenly across the available storage space in the hash table

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Hash table: trace algorithm 1

Advanced: Challenge 2 

A hash function is an algorithm that converts a hash key to a hash value. A simple example of a hash function is represented by the pseudocode shown below:

```
1 FUNCTION hash_it(hash_key, number_of_slots)
2     hash_value = hash_key MOD number_of_slots
3     RETURN hash_value
4 ENDFUNCTION
```

A hash table has 48 slots (buckets) and hash keys are five digit numbers such as those shown in the list below:

- 31971
- 43219
- 56342
- 96756

If the keys listed are hashed using the example of the simple hashing algorithm (shown above), what hash values will be generated?

- ☐

Key	Hash value
31971	1
43219	4
56342	8
96756	7
- ☐

Key	Hash value
31971	51
43219	67
56342	86
96756	84
- ☐

Key	Hash value
31971	666

43219	900
56342	1173
96756	2015



Key	Hash value
31971	3
43219	19
56342	38
96756	36

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Hash table: trace algorithm 2

Advanced: Challenge 2  

Consider an alphanumeric key that is made up of any five letters or numbers. An example would be **AZ3FC**.

A hash function calculates hash values as follows:

- each character in the key is converted to its ASCII decimal numeric code **including numeric values** (i.e. A will be converted to 65, 1 will be converted to 49)
- the numeric codes are summed to give a total
- the total is divided by the number of slots in the hash table **using modular division**
- the result is the hash value

If a hash table has 55 slots (buckets) what hash value will be generated for the key value **AZ3FC**?

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Hash table: dealing with collisions

Advanced: Practice 2  

A delivery company stores parcel IDs (keys) in a hash table with 10 slots (buckets).

The hash function that is used to determine the index is:

$$\text{index} = \text{key} \text{ MOD } 10$$

where MOD is the modulo operator.

Any collisions are resolved using **linear probing**.

The following keys have already been inserted:

12, 22 and 35

The hash table now looks like this:

Index	Key
0	Empty
1	Empty
2	12
3	22
4	Empty
5	35
6	Empty
7	Empty
8	Empty
9	Empty

Part A Insert key 32

At which index position will the key 32 be stored?

- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

Part B Insert key 42

After adding 32, key 42 is added. At which index position would this key be stored?

Part C Insert four more keys

Continue inserting the following keys into the hash table, in the order shown.

17, 15, 18 and 1

Any collisions should be resolved using linear probing.

Complete the final hash table with the correct positions of the keys:

Index	Key
0	Empty
1	
2	12
3	22
4	32
5	35
6	42
7	
8	
9	

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Algorithms: efficiency

Core: Practice 2  

There are often many different algorithms that can accomplish the same task.

Which of the following characteristics should you consider when comparing the efficiency of two sorting algorithms?

- ☐ The number of comparisons
- ☐ The number of subroutines
- ☐ The number of constants

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Binary search tree: trace

Advanced: Challenge 2 ■■

A wildfowl trust tags the birds that visit over the year with a unique number in order to keep track of them. The details of the birds are stored in a **binary search tree** for efficient searching.

The numbers from the **tags of 13 birds** have been entered into the tree, and these are shown in the diagram in **Figure 1** below.

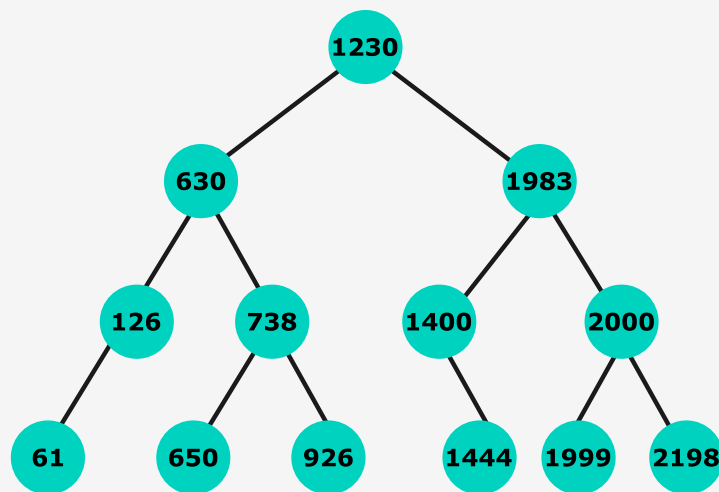


Figure 1: A binary search tree containing data from the tags of 13 birds.

Part A Search for 1999

In what order will the nodes of the binary search tree be visited when the bird with tag number **1999** is searched for?

Select and drag **only the relevant numbers** into the correct order. Not all numbers need to be used.

Available items

126
630
738
1230
1400
1983
1999
2000

Part B Search for 750

A bird has been found with the tag number **750**. In what order will the nodes of the binary search tree be visited when searching for the tag **750** to find out whether (or not) the tag exists in the binary search tree?

Select and drag **only the relevant numbers** into the correct order. Not all numbers need to be used.

Available items

126
630
650
738
926
1230
1400
1983
2000

Part C Maximum nodes examined

In the example provided, what is the **maximum** number of nodes of the binary search tree that will need to be examined to find any bird?

Part D Time complexity - balanced tree

The tree illustrated in **Figure 1** is an example of a **balanced tree**. For a balanced tree, with n nodes, what is the worst-case time complexity in Big O notation of the binary tree search algorithm?

- ☐ $O(n^2)$
- ☐ $O(n)$
- ☐ $O(\log n)$
- ☐ $O(n \log n)$

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Big O: order of complexity

Advanced: Practice 2  

Big O notation is a standard way to express an algorithm's time or space complexity. It allows us to understand how the performance of an algorithm scales as the size of the input grows.

Part A

Choose the Big O expression with the **lowest** complexity from the options provided.

- ☐ $\mathcal{O}(n!)$
- ☐ $\mathcal{O}(n)$
- ☐ $\mathcal{O}(\log n)$

Part B

Choose the Big O expression with the **highest** complexity from the options provided.

- ☐ $\mathcal{O}(n!)$
- ☐ $\mathcal{O}(\log n)$
- ☐ $\mathcal{O}(n)$

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Big O: rank complexities

Advanced: Challenge 2  

Put the following Big O expressions into the order of increasing complexity.

Available items

$O(n!)$

$O(1)$

$O(\log n)$

$O(n \log n)$

$O(2^n)$

$O(n^2)$

$O(n)$

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Big O: determine expression 1

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The number of operations (steps) of an algorithm is given below, where n is the size of the input:

$$4n^3 + 3n \log_2 n + 273$$

How would the time complexity of this algorithm be expressed in **Big O notation**?

Type your answer in the format $\mathcal{O}(n)$, replacing n with the relevant term. If you want to use an exponent, use the $\wedge$ symbol, for example, $\mathcal{O}(n^2)$.

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Big O: determine expression 2

Advanced: Challenge 2  

The number of operations (steps) of an algorithm is given below, where n is the size of the input:

$$5n^2 + \log_2 n + 2^n + 18$$

How would the time complexity of this algorithm be expressed in **Big O notation**?

Type your answer in the format $\mathcal{O}(n)$, replacing n with the relevant term. If you want to use an exponent, use the $\wedge$ symbol, for example, $\mathcal{O}(n^2)$.

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